Week 3 reading guide and worked examples

Weekly overview

Required learning outcomes

By the end of this week, you should be able to:

- **M5:** determine span, linear independence, bases and dimension; compute eigenvalues and eigenvectors; and diagonalise a simple real matrix when diagonalisation is possible;
- **S3:** use probability mass functions (PMFs), probability density functions (PDFs) and cumulative distribution functions (CDFs), and work with the required discrete and continuous distributions under the course parameterisations; and
- **S4:** calculate and interpret expectations, variances, covariances and correlations, including under linear transformations and by using indicator variables.

Prerequisites and retrieval

You will use matrix operations, determinants, rank and elimination from `M4`; algebra and logarithms from `M2`; probability rules from `S1`; and conditional probability from `S2`. Question 10 contains three explicit earlier-outcome retrieval parts on `M4`, `S2` and `M2`.

Before beginning, check that you can multiply a matrix by a column vector, row-reduce a small augmented matrix, solve a quadratic equation and apply the multiplication rule for probabilities.

Student materials

1. Follow the exact external sections and skip instructions in the reading guide.
2. Within the registered reading-and-example block, study and reproduce the original worked examples.
3. Attempt every part of the exercise sheet without full solutions.
4. Consult the progressive hints only after a serious attempt.
5. Mark from the complete solutions, record the first cause of each error by outcome ID, and reattempt missed work after two or three days.
6. Finish with the 25-minute self-test under closed-note conditions.

The external readings remain on their authors' or institutions' sites. Use only the listed sections. No video, applet, code or specialist software is required.

Suggested study sequence

1. Retrieve the prerequisite methods and write `M5`, `S3` and `S4` at the top of your error log.
2. Complete the vector-space and eigenstructure reading. Reproduce Worked Examples 1 and 2 from a blank page.
3. Complete the random-variable and distribution readings. Build one distribution card that states the course parameterisation and support for every required distribution.
4. Complete the moments reading. Reproduce Worked Examples 3–5, checking the distinction between a PMF, PDF and CDF and between covariance and correlation.
5. Attempt Questions 1–9, then complete the Question 10 retrieval block without earlier notes.
6. Use hints one at a time only when a written attempt has reached a specific impasse.
7. Mark the sheet, record the first cause of each error and take the self-test without notes.
8. Reattempt every incorrect or incomplete part from a blank page two or three days later.

Workload and required outputs

The times below are approximate planning estimates and have not been verified by students. The first live cohort will be invited to report actual component times so that the plan can be recalibrated.

The approved resource-map estimates for Week 3 already include time for worked-example study. The institutional worked examples therefore use time inside the 335-minute reading-and-example block; they are not an additional core component.

Component	Target time	Required output
Exact external reading and worked-example reproduction	5 h 35 min	Brief original notes, one distribution card, and five examples reproduced or checked from a blank page
Reading-to-example planning check	5 min	Mark the examples that require a later reattempt
Week 3 exercise sheet	4 h	All 17 scorable parts attempted and marked
Self-test	25 min	All four self-test parts attempted and marked
Error logging and first spaced reattempt	25 min	Outcome-labelled error entries and a first blank-page reattempt of missed work
Core total	**10 h 30 min**	Completed Week 3 study cycle

The exercise sheet contains 10 numbered questions, 17 scorable parts, 100 marks and a 240-minute target. Across marks, it contains 33% `FOUNDATION`, 57% `STANDARD` and 10% `STRETCH` work. Parts classified `EXPLAIN`, `DERIVE` or `SYNTHESISE` carry 45% of the marks.

End-of-week check

Before moving on, make sure that you can state why a proposed spanning set is or is not a basis, when a matrix is diagonalisable over the real numbers, and why a density value is not itself a probability. You should also be able to state the support and parameterisation of each required distribution without consulting the source.

Required reading route

How to use this route

Complete the four required written selections below in order. Open only the named sections and obey every skip instruction. The registered total is 335 minutes and already includes worked-example study. Within each resource allocation, close the source periodically and reproduce the corresponding institutional example in worked examples. After this 335-minute block, use the separately budgeted five-minute planning check to identify which examples need a later blank-page reattempt.

Do not reproduce external prose, exercise prompts or solutions in shared notes. Write a brief original summary and use the course's notation and parameterisations.

1. Span, independence, basis and eigenstructure (`M5`)

Resource `MATH-M5-HEF-REQ`; source `HEFFERON_LA`. In Jim Hefferon, *Linear Algebra*, read:

- Chapter Two I.2, "Subspaces and Spanning Sets";
- Chapter Two II.1, "Definition and Examples" under Linear Independence;
- Chapter Two III.1, "Basis";
- Chapter Two III.2, "Dimension";
- Chapter Five II.2, "Diagonalizability"; and
- Chapter Five II.3, "Eigenvalues and Eigenvectors".

Skip: complex-root examples, proof-heavy extensions, starred subsections, Jordan form, nilpotence, chapter-end applications, the Sage lab and videos. Use real vectors and real matrices only. Do not treat the whole book or the separately linked answer book as assigned.

Registered reading-and-example time: 135 minutes. Use part of this allocation to reproduce Worked Examples 1 and 2.

2. PMFs, PDFs, CDFs and the core distributions (`S3`)

Resource `STAT-S3-REQ-MIT`; source `MIT_1805`. On Jeremy Orloff and Jonathan Bloom's MIT OpenCourseWare 18.05 classes and written-materials page, read:

- Class 4 Reading 4a §§2.1–3.5;
- Class 5 Reading 5b §§4–4.5; and
- Class 5 Reading 5c §§2–5.1.

Skip: Reading 4a §§3.6–3.7, including the applet; Reading 5c §5.2, which uses R; Reading 5c §6 on Pareto and other distributions; Problem Set 3 Problems 2, 5(a–b) and 6(a), which require R; and every linked video, interactive question or software activity.

Use the course convention for the geometric distribution: it counts trials up to and including the first success and has support $\{1, 2, ...\}$. For a normal distribution, the second parameter in $\mathcal{N}(\mu, \sigma^2)$ is the variance.

Registered reading-and-example time: 90 minutes. Use part of this allocation to reproduce Worked Example 3 and build the distribution card.

3. The Poisson distribution (`S3`)

Resource `STAT-S3-REQ-PISHRO-POISSON`; source `PISHRO_NIK`. In Hossein Pishro-Nik, *Introduction to Probability, Statistics, and Random Processes*, read only the "Poisson Distribution" subsection of §3.1.5.

Skip: every other §3.1.5 distribution subsection, including the negative-binomial/Pascal and hypergeometric distributions. The Poisson approximation to the binomial is enrichment, not core work. Do not copy or adapt the page: the approved rights record requires external-link-only use for this exact Poisson subsection.

Registered reading-and-example time: 20 minutes.

4. Expectation, variance, covariance and correlation (`S4`)

Resource `STAT-S4-REQ-MIT`; source `MIT_1805`. On the same MIT OpenCourseWare 18.05 classes and written-materials page, read:

- Class 4 Reading 4b §§2.1–2.4;
- Class 5 Reading 5a §§2–4;
- Class 6 Reading 6a §§3–4.1; and
- Class 7 Reading 7b §§2–3.1.

Skip: Class 6 Reading 6a §§5–6; Class 7 Reading 7b §§3.2–4 on the bivariate normal, overlapping uniforms and proofs; and every linked video or software activity.

Registered reading-and-example time: 90 minutes. Use part of this allocation to reproduce Worked Examples 4 and 5.

Required reading output

After each selection, close the source and write a short original summary. By the end of the route, your notes should contain:

- one span/independence/basis decision with its rank or coefficient check;
- one complete real diagonalisation, including $\mathbf{A} = \mathbf{PDP}^{-1}$;

- one distribution card giving the support and parameterisation of the Bernoulli, binomial, geometric, Poisson, continuous uniform, exponential and normal distributions;
- one PMF or PDF normalisation and CDF calculation; and
- the expectation, variance, covariance, correlation and linear- transformation formulae with their conditions.

Worked examples

These are original institutional notes and examples for `M5`, `S3` and `S4`. Their study time is included in the 335-minute registered reading-and-example block. Cover each **Solution**, follow the **Method** and reproduce the calculation from a blank page. All vectors are column vectors.

Definitions and conventions

For vectors $\mathbf{v}_1, \dots, \mathbf{v}_k \in \mathbb{R}^n$, their span is

$$\operatorname{span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\} := \left\{ \sum_{i=1}^{k} a_i \mathbf{v}_i : a_1, \dots, a_k \in \mathbb{R} \right\}.$$

The vectors are linearly independent when

$$\sum_{i=1}^{k} a_i \mathbf{v}_i = \mathbf{0} \quad \Longrightarrow \quad a_1 = \cdots = a_k = 0.$$

A basis of a subspace is a linearly independent spanning set. The dimension is the number of vectors in any basis of that subspace.

For a square real matrix $\mathbf{A}$, a non-zero vector $\mathbf{v}$ is an eigenvector associated with λ when

$$\mathbf{A}\mathbf{v} = \lambda \mathbf{v}.$$

An $n \times n$ matrix is diagonalisable over $\mathbb{R}$ when it has n linearly independent real eigenvectors. If these eigenvectors are the columns of $\mathbf{P}$, then

$$\mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^{-1},$$

where $\mathbf{D}$ contains the corresponding eigenvalues on its diagonal.

For probability, use $p_X(x)$ for a PMF, $f_X(x)$ for a PDF and $F_X(x) = \mathbb{P}(X \le x)$ for a CDF. A PMF sums to one over its support. A PDF is non-negative and integrates to one over its support. A density value $f_X(x)$ is not a point probability: for a continuous random variable, $\mathbb{P}(X = x) = 0$.

The required distributions use these course parameterisations:

Distribution	Support and parameterisation	Mean	Variance
Bernoulli(p)	$\{0, 1\}$, with $\mathbb{P}(X = 1) = p$ and $0 \le p \le 1$	p	$p(1-p)$
Binomial(n, p)	$\{0, \dots, n\}$, successes in n independent Bernoulli trials, with $n \in \{1, 2, \dots\}$ and $0 \le p \le 1$	np	$np(1-p)$

Distribution	Support and parameterisation	Mean	Variance
Geometric(p)	$\{1, 2, ...\}$, trials through the first success, with $0 < p \leq 1$	$1/p$	$(1-p)/p^2$
Poisson(λ)	$\{0, 1, 2, ...\}$, $\lambda > 0$	λ	λ
Uniform(a, b)	$[a, b]$, density $1/(b-a)$, $a < b$	$(a+b)/2$	$(b-a)^2/12$
Exponential(λ)	$[0, \infty)$, rate $\lambda > 0$	$1/\lambda$	$1/\lambda^2$
$\mathcal{N}(\mu, \sigma^2)$	$\mathbb{R}$, variance $\sigma^2 > 0$	μ	σ^2

Example 1 — span, independence and coordinates (`M5`)

Let

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \qquad \mathbf{v}_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \qquad \mathbf{v}_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}.$$

Show that $\mathcal{B} = (\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$ is a basis of $\mathbb{R}^3$, then find the coordinates of $\mathbf{x} = [6, 5, 3]^\mathsf{T}$ relative to $\mathcal{B}$.

Method. Put the vectors into the columns of a matrix. A non-zero determinant establishes independence; three independent vectors in $\mathbb{R}^3$ form a basis. Solve the coefficient equation for the coordinates.

Solution. Let

$$\mathbf{P} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}.$$

Expansion along the first row gives

$$\det(\mathbf{P}) = 1 \neq 0.$$

Thus the columns are linearly independent. Since there are three of them in $\mathbb{R}^3$, they also span $\mathbb{R}^3$, so $\mathcal{B}$ is a basis.

Write

$$a\mathbf{v}_1 + b\mathbf{v}_2 + c\mathbf{v}_3 = \begin{bmatrix} 6 \\ 5 \\ 3 \end{bmatrix}.$$

The coefficient equations are

$$a + b + c = 6, \qquad b + c = 5, \qquad c = 3.$$

The third equation gives $c = 3$; the second then gives $b = 2$, and the first gives $a = 1$. Therefore,

$$[\mathbf{x}]_{\mathcal{B}} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}.$$

Interpretation. The same coefficient calculation both represents $\mathbf{x}$ as a linear combination and gives its coordinates in the ordered basis. Reordering the basis would reorder the coordinate vector.

Example 2 — eigenvalues and diagonalisation (`M5`)

Let

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}.$$

Find its eigenvalues and eigenvectors and diagonalise it over $\mathbb{R}$.

Method. Solve the characteristic equation $\det(\mathbf{A} - \lambda \mathbf{I}_2) = 0$. For each eigenvalue, solve $(\mathbf{A} - \lambda \mathbf{I}_2)\mathbf{v} = \mathbf{0}$. Put independent eigenvectors in the columns of $\mathbf{P}$ in the same order as their eigenvalues in $\mathbf{D}$.

Solution. The characteristic polynomial is

$$\begin{aligned} \det(\mathbf{A} - \lambda \mathbf{I}_2) &= \begin{vmatrix} 1-\lambda & 2 \\ 2 & 1-\lambda \end{vmatrix} \\ &= (1-\lambda)^2 - 4 \\ &= \lambda^2 - 2\lambda - 3 \\ &= (\lambda - 3)(\lambda + 1). \end{aligned}$$

Thus the eigenvalues are 3 and -1.

For $\lambda = 3$,

$$(\mathbf{A} - 3\mathbf{I}_2)\mathbf{v} = \begin{bmatrix} -2 & 2 \\ 2 & -2 \end{bmatrix} \mathbf{v} = \mathbf{0},$$

so one eigenvector is $\mathbf{v}_1 = [1, 1]^{\mathsf{T}}$. For $\lambda = -1$,

$$(\mathbf{A} + \mathbf{I}_2)\mathbf{v} = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix} \mathbf{v} = \mathbf{0},$$

so one eigenvector is $\mathbf{v}_2 = [1, -1]^{\mathsf{T}}$.

These eigenvectors are independent because

$$\det \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = -2 \neq 0.$$

Therefore, with

$$\mathbf{P} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \qquad \mathbf{D} = \begin{bmatrix} 3 & 0 \\ 0 & -1 \end{bmatrix},$$

the matrix is diagonalisable and $\mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^{-1}$. Indeed,

$$\mathbf{P}^{-1} = \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{bmatrix}.$$

Interpretation. The columns of $\mathbf{P}$ and the diagonal entries of $\mathbf{D}$ must remain paired. Distinct real eigenvalues guarantee independent eigenvectors in this 2×2 case.

Example 3 — from a PMF to a CDF and moments (`S3`, `S4`)

Let X have support $\{0, 1, 2\}$ and PMF

$$p_X(x) = c(2x + 1), \qquad x \in \{0, 1, 2\}.$$

Find c, the CDF, the mean and the variance.

Method. Normalise the PMF, accumulate its masses to form the CDF, then calculate $\mathbb{E}[X]$ and $\mathbb{E}[X^2]$.

Solution. Normalisation requires

$$c(1 + 3 + 5) = 9c = 1,$$

so $c = 1/9$. The CDF is

$$F_X(x) = \begin{cases} 0, & x < 0, \\ 1/9, & 0 \le x < 1, \\ 4/9, & 1 \le x < 2, \\ 1, & x \ge 2. \end{cases}$$

The first two moments needed for the variance are

$$\mathbb{E}[X] = 0\frac{1}{9} + 1\frac{3}{9} + 2\frac{5}{9} = \frac{13}{9}$$

and

$$\mathbb{E}[X^2] = 0^2\frac{1}{9} + 1^2\frac{3}{9} + 2^2\frac{5}{9} = \frac{23}{9}.$$

Therefore,

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = \frac{23}{9} - \left(\frac{13}{9}\right)^2 = \frac{38}{81}.$$

Interpretation. A discrete CDF is right-continuous and jumps by the PMF mass at each support point.

Example 4 — a continuous distribution and a linear transformation (`S3`, `S4`)

Let $T \sim$ Exponential(3), where 3 is the rate. Find $\mathbb{P}(T > 2)$, then find the mean and variance of $Y = 2T + 4$.

Method. State the support and exponential parameterisation. Integrate the density for the tail, then use the linear-transformation rules for moments.

Solution. The support is $t \ge 0$, and the density is

$$f_T(t) = 3e^{-3t}, \qquad t \ge 0.$$

Thus

$$\mathbb{P}(T > 2) = \int_2^\infty 3e^{-3t}\,\mathrm{d}t = e^{-6}.$$

Since the rate is 3,

$$\mathbb{E}[T] = \frac{1}{3}, \qquad \mathrm{Var}(T) = \frac{1}{9}.$$

For $Y = 2T + 4$,

$$\mathbb{E}[Y] = 2\mathbb{E}[T] + 4 = \frac{14}{3},$$

and

$$\mathrm{Var}(Y) = 2^2\,\mathrm{Var}(T) = \frac{4}{9}.$$

Interpretation. Adding a constant changes the mean but not the variance. Multiplying by 2 multiplies variance by 2^2.

Example 5 — indicators, covariance and dependence (S4)

Two cards are drawn without replacement from six cards, of which two are red. Let I_1 indicate that the first card is red and I_2 indicate that the second card is red. Let $R = I_1 + I_2$, the number of red cards drawn. Find $\mathbb{E}[R]$, $\mathrm{Var}(R)$ and $\mathrm{Corr}(I_1, I_2)$.

Method. Each indicator is Bernoulli. Use linearity of expectation and include the covariance term in the variance of the sum.

Solution. By symmetry,

$$\mathbb{P}(I_1 = 1) = \mathbb{P}(I_2 = 1) = \frac{1}{3}.$$

Therefore,

$$\mathbb{E}[I_1] = \mathbb{E}[I_2] = \frac{1}{3}, \qquad \mathrm{Var}(I_1) = \mathrm{Var}(I_2) = \frac{2}{9}.$$

The probability that both cards are red is

$$\mathbb{P}(I_1 = 1, I_2 = 1) = \frac{1}{3}\frac{1}{5} = \frac{1}{15}.$$

Hence

$$\mathrm{Cov}(I_1, I_2) = \mathbb{E}[I_1 I_2] - \mathbb{E}[I_1]\mathbb{E}[I_2] = \frac{1}{15} - \frac{1}{9} = -\frac{2}{45}.$$

It follows that

$$\mathbb{E}[R] = \frac{1}{3} + \frac{1}{3} = \frac{2}{3}$$

and

$$\mathrm{Var}(R) = \frac{2}{9} + \frac{2}{9} + 2\left(-\frac{2}{45}\right) = \frac{16}{45}.$$

Finally,

$$\mathrm{Corr}(I_1, I_2) = \frac{-2/45}{(\sqrt{2}/3)(\sqrt{2}/3)} = -\frac{1}{5}.$$

Interpretation. The negative covariance reflects sampling without replacement: a red card on the first draw makes a red card less likely on the second draw.

Reproduction check

Before attempting the exercise sheet, reproduce:

1. the coefficient or determinant check that establishes a basis;
2. one complete eigenvector calculation and the pairing of $\mathbf{P}$ with $\mathbf{D}$;
3. one piecewise CDF;
4. one linear transformation of a mean and variance; and
5. the covariance term in the variance of an indicator sum.